Do driver mutations add prognostic information beyond stage in lung adenocarcinoma?

A clinical–genomic survival analysis of TCGA-LUAD with matched RNA expression

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Abstract

Background. *EGFR*, *KRAS*, *TP53* and *STK11* are among the most frequently altered genes in lung adenocarcinoma and guide treatment, but how much prognostic information their mutation status carries once stage is known is less clear. **Methods.** I analysed 295 patients (109 deaths) from the TCGA-LUAD PanCancer Atlas cohort. A Cox model with age, sex and pathological stage (M0) was compared with the same model plus four binary mutation indicators (M1), using a likelihood-ratio test and Harrell's C-index corrected for optimism with 200 bootstrap resamples. In 293 patients with matched RNA-seq, I compared the expression of each gene between mutated and non-mutated tumours. **Results.** Stage dominated both models. Adding the mutations raised the apparent C-index from 0.668 to 0.688, but the gain fell to 0.006 after optimism correction (0.654 vs 0.660) and the joint contribution of the four genes was not significant (likelihood-ratio $\chi^2 = 7.23$, 4 d.f., $p = 0.12$). At the transcript level all four genes differed by mutation status: expression was higher in *EGFR*- and *KRAS*-mutated tumours and lower in *TP53*- and *STK11*-mutated tumours (all $q < 0.01$). **Conclusion.** In this cohort, presence or absence of a mutation in these four genes adds little to stage for ranking patients by survival, even though the same mutations leave a clear mark on expression. A binary mutation flag is probably too coarse a summary of the genomic layer for prognosis.

1 Introduction

Lung adenocarcinoma (LUAD) is the most common histological subtype of non-small-cell lung cancer. Its genomic landscape was mapped in detail by The Cancer Genome Atlas (TCGA), which identified recurrent alterations in *TP53*, *KRAS*, *EGFR* and *STK11*, among others [1]. These genes matter clinically mainly as predictors of treatment benefit: *EGFR* and *KRAS* mutations define targetable subgroups, and *STK11* loss has been linked to resistance to immune checkpoint blockade in *KRAS*-mutant disease [2]. Their value as *prognostic* markers, independent of treatment, is a separate question, and pathological stage remains the reference predictor of survival.

This project asks a narrow question that can be answered with public data: once age, sex and stage are accounted for, does knowing whether these four genes are mutated improve the ability to rank patients by overall survival? A second, descriptive analysis links the DNA and RNA layers by asking whether the mutation status of each gene is reflected in its own transcript abundance. I chose a small, pre-specified model over a high-dimensional one on purpose: with 109 events, the data support about ten coefficients, and a model of this size can be validated and interpreted without feature selection.

2 Methods

2.1 Data and cohort

Clinical, somatic mutation and RNA-seq data were obtained from the TCGA-LUAD PanCancer Atlas study on cBioPortal (`luad_tcga_pan_can_atlas_2018`) [3–5]. The study contains 566 patients. Of these, 489 had complete age, sex, pathological stage and overall survival (OS) with positive follow-up time. To keep the time origin of OS aligned with the specimen that was sequenced, I retained the 295 patients whose sample acquisition was recorded at day zero in the study timeline. Each patient contributed one primary tumour sample. Stage sub-categories were collapsed into stages I–IV.

A gene was coded as mutated when the sample carried at least one variant with `FILTER = PASS` classified as missense, nonsense, nonstop, frameshift or in-frame insertion or deletion, splice-site or translation-start-site. All samples were profiled by whole-exome sequencing; a gene with no qualifying variant was coded as non-mutated. No values were imputed.

2.2 Survival models

The endpoint was overall survival, in months. Two nested Cox proportional-hazards models were fitted without penalisation (Efron method for ties):

- **M0, clinical:** age (per 10 years), sex, and stage as a categorical variable (5 coefficients);
- **M1, clinical + mutations:** M0 plus the four mutation indicators (9 coefficients, 12.1 events per coefficient).

The joint contribution of the four indicators was tested with a likelihood-ratio test on 4 degrees of freedom. Discrimination was measured with Harrell's C-index [6]. Because both models are evaluated on the data used to fit them, I estimated optimism with the bootstrap [7]: in each of 200 resamples (drawn with replacement, same resample for both models) the models were refitted and scored on the resample and on the original cohort, and the mean difference was subtracted from the apparent C-index. The same correction was applied to the difference $\Delta C = C_{\mathrm{M1}} - C_{\mathrm{M0}}$. The proportional-hazards assumption was examined with tests based on scaled Schoenfeld residuals [8] under rank and Kaplan–Meier time transformations. Unadjusted survival was described with Kaplan–Meier curves and log-rank tests; median follow-up was estimated with the reverse Kaplan–Meier method.

2.3 Mutation–expression analysis

Expression values were batch-normalised RSEM estimates [9] from the same study, transformed as $\log_2(\mathrm{RSEM} + 1)$. Samples were matched on the full sample identifier; 293 of the 295 patients had RNA-seq. For each of the four genes, expression was compared between mutated and non-mutated tumours with a two-sided Mann–Whitney U test. Effect size is reported as the difference in medians and as the rank-biserial correlation. The four p-values were adjusted with the Benjamini–Hochberg procedure [10] and are reported as q-values. The genes were fixed before the expression data were examined.

2.4 Software

Analyses were run in Python (3.12 and 3.13) with lifelines 0.30.3 [11], SciPy, pandas and matplotlib. The bootstrap uses a fixed seed (42). Code, the analysis table and all outputs are in the project repository.

3 Results

3.1 Cohort

The cohort comprised 295 patients with a median age of 66 years (Table 1). Just over half had stage I disease and 19 had stage IV. During a median follow-up of 30.8 months, 109 patients died; median OS was 49.3 months. *TP53* was mutated in 47.5% of tumours, *KRAS* in 29.2%, *STK11* in 13.2% and *EGFR* in 12.9%, in line with frequencies reported for this tumour type [1].

Table 1. Characteristics of the analysis cohort ($n = 295$).

	Patients	Deaths		Patients	Deaths
All patients	295	109	**Mutated gene**		
Age, median (IQR), years	66 (59–72)		*TP53*	140 (47.5%)	52
Female sex	159 (53.9%)		*KRAS*	86 (29.2%)	33
Pathological stage			*STK11*	39 (13.2%)	16
I	159 (53.9%)	39	*EGFR*	38 (12.9%)	17
II	69 (23.4%)	31	**Follow-up**		
III	48 (16.3%)	29	Median follow-up, months		30.8
IV	19 (6.4%)	10	Median OS, months		49.3

3.2 Survival by stage and mutation status

Survival separated clearly by stage (log-rank $p < 0.001$; Fig. 1a). Unadjusted survival did not differ significantly by *KRAS* status (log-rank $p = 0.16$), although the curves diverged during the first two years and crossed after about four (Fig. 1b).

In the clinical model, stage II, III and IV were each associated with a two- to three-fold higher hazard of death than stage I (Table 2, Fig. 2a). The stage IV estimate was lower than the stage III estimate, with a wide interval reflecting only 19 patients. In M1 all four mutation indicators had hazard ratios above 1, between 1.31 and 1.56, but every confidence interval included 1. The largest estimates were for *KRAS* (HR 1.56, 95% CI 1.00–2.43) and *STK11* (HR 1.56, 0.89–2.74). Stage coefficients changed little when the

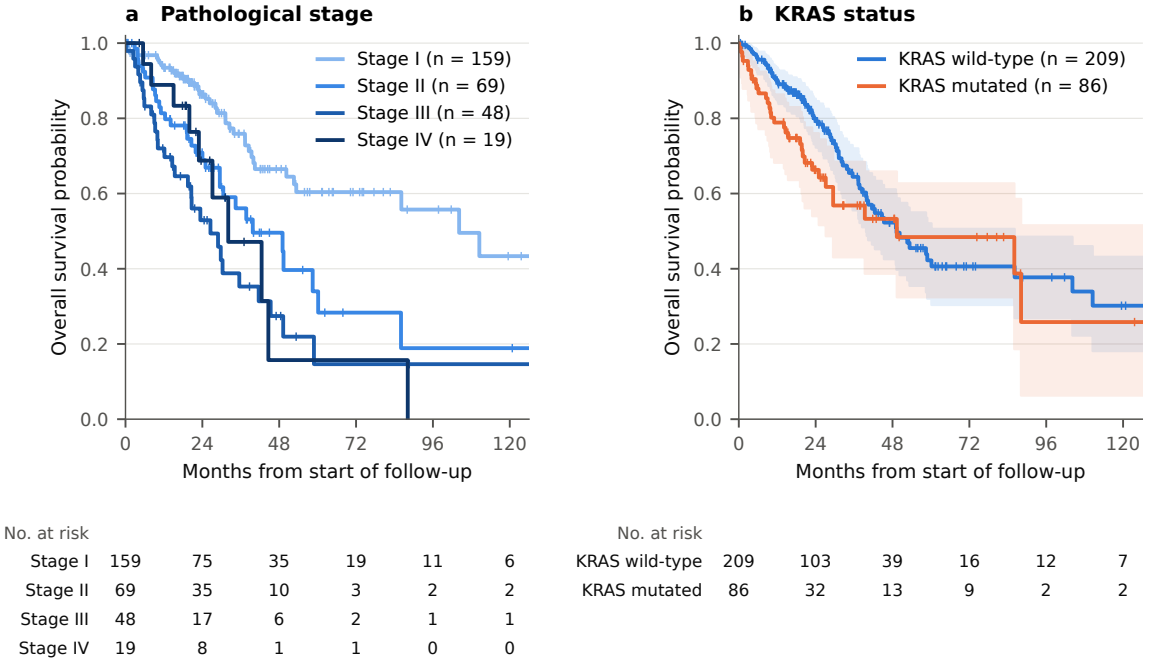


Figure 1. Kaplan–Meier estimates of overall survival by pathological stage (a) and by *KRAS* mutation status (b). Tick marks indicate censoring; shaded bands in (b) are 95% confidence intervals. The axis is truncated at 126 months; fewer than 10 patients remain at risk beyond 120 months.

Table 2. Adjusted hazard ratios for death from the two Cox models.

	M0, clinical		M1, clinical + mutations	
Covariate	HR (95% CI)	p	HR (95% CI)	p
Age, per 10 years	1.15 (0.94–1.40)	0.17	1.22 (0.99–1.50)	0.058
Male sex	1.16 (0.79–1.70)	0.46	1.13 (0.76–1.70)	0.54
Stage II vs I	2.22 (1.38–3.57)	< 0.001	2.08 (1.29–3.36)	0.003
Stage III vs I	3.49 (2.15–5.66)	< 0.001	3.40 (2.06–5.61)	< 0.001
Stage IV vs I	2.75 (1.35–5.60)	0.006	2.52 (1.22–5.19)	0.013
EGFR mutated			1.31 (0.74–2.30)	0.35
KRAS mutated			1.56 (1.00–2.43)	0.052
TP53 mutated			1.31 (0.88–1.94)	0.19
STK11 mutated			1.56 (0.89–2.74)	0.12

mutations were added.

3.3 Added value of the mutation indicators

The four indicators together did not significantly improve model fit (likelihood-ratio $\chi^2 = 7.23$, 4 d.f., $p = 0.12$), and the partial AIC was marginally lower for the clinical model (1036.9 vs 1037.7). The apparent C-index rose from 0.668 to 0.688 (Table 3). M1, with nearly twice as many coefficients, was also more optimistic: its mean optimism was 0.028 against 0.015 for M0. After correction the two models were almost indistinguishable, 0.654 and 0.660, a difference of 0.006. In other words, about two thirds of the apparent gain from the mutations is what one would expect from fitting four extra parameters to this sample (Fig. 2b). All 200 bootstrap fits converged without warnings.

Table 3. Discrimination of the two models. Optimism is the mean over 200 bootstrap resamples.

	M0	M1	M1 − M0
Apparent C-index	0.668	0.688	+0.019
Mean optimism	0.015	0.028	+0.013
Optimism-corrected C-index	0.654	0.660	+0.006

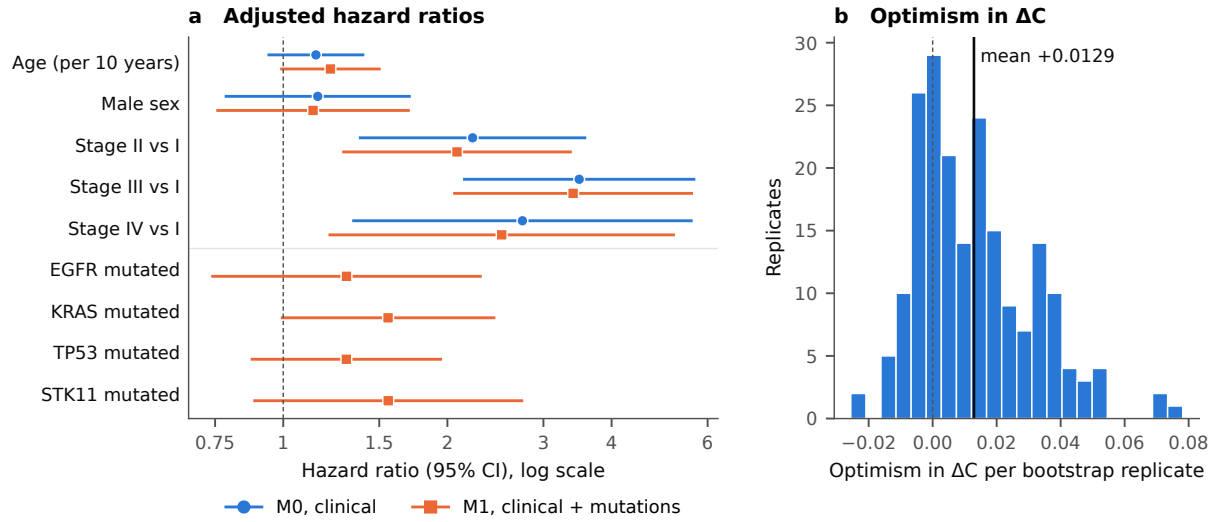


Figure 2. (a) Hazard ratios and 95% confidence intervals from the clinical model (M0) and the model with mutation indicators (M1). (b) Distribution over 200 bootstrap replicates of the optimism in the C-index difference between M1 and M0; the solid line is the mean, which is subtracted from the apparent difference.

3.4 Proportional-hazards diagnostics

The Schoenfeld tests flagged sex in M0 ($p = 0.049$ and 0.041 under the rank and Kaplan–Meier transformations) and *KRAS* in M1 ($p = 0.008$ and 0.028); no other covariate fell below 0.05 in either model. The *KRAS* signal agrees with the crossing curves in Fig. 1b: the excess risk in *KRAS*-mutated patients appears concentrated early in follow-up. The hazard ratio of 1.56 should therefore be read as an average over follow-up and not as a constant effect.

3.5 Mutation status and expression

In the 293 patients with matched RNA-seq (108 deaths), the expression of every gene differed according to its own mutation status (Table 4, Fig. 3). *EGFR*-mutated tumours expressed more *EGFR* (median difference $+1.17$ on the $\log_2$ scale, roughly 2.3-fold) and *KRAS*-mutated tumours more *KRAS* ($+0.45$). The two tumour suppressors went in the opposite direction: expression was lower in *STK11*-mutated (-0.45) and *TP53*-mutated tumours (-0.21). The distributions overlap substantially in all four comparisons, most of all for *TP53*.

Table 4. Expression of each gene, $\log_2(\text{RSEM} + 1)$, by its own mutation status in 293 matched samples.

		Median (IQR)				
Gene	n, WT / mut.	Non-mutated	Mutated	Difference	Rank-biserial r	q
EGFR	255 / 38	9.75 (8.95–10.56)	10.93 (10.18–11.78)	$+1.17$	$+0.51$	6.3×10^{-7}
KRAS	208 / 85	10.29 (9.93–10.65)	10.75 (10.41–11.15)	$+0.45$	$+0.49$	2.0×10^{-10}
TP53	154 / 139	10.47 (10.16–10.87)	10.25 (9.59–10.81)	-0.21	-0.20	0.003
STK11	255 / 38	9.78 (9.32–10.10)	9.33 (8.78–9.71)	-0.45	-0.33	0.002

4 Discussion

The main result is negative, and I think it is informative. Stage carries most of the prognostic signal in this cohort, and whether *EGFR*, *KRAS*, *TP53* or *STK11* is mutated adds very little to it: an optimism-corrected gain of 0.006 in the C-index, with no significant improvement in fit. The uncorrected figures would have told a more flattering story. A reader shown only the apparent C-index (0.688 vs 0.668) might conclude that the genomic layer helps; the bootstrap shows that most of that gain comes from model size.

This does not mean that the genes are irrelevant to outcome. All four hazard ratios point towards higher risk, and the estimates for *KRAS* and *STK11* are not small; with 109 deaths the study cannot distinguish a hazard ratio of 1.5 from no effect with any confidence. The time pattern for *KRAS* is also

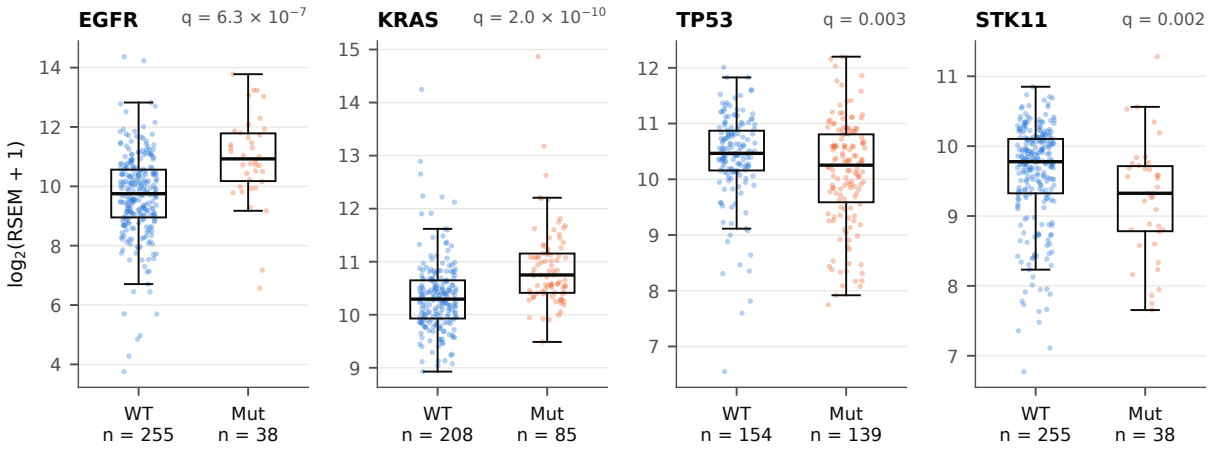


Figure 3. Expression of each gene in tumours without (WT) and with (Mut) a qualifying mutation in that gene. Boxes show the median and interquartile range, whiskers extend to 1.5 times the interquartile range, and each point is one patient. q: Benjamini–Hochberg adjusted p-value across the four tests.

worth following up. An early excess risk that fades is what one would see if *KRAS*-mutated tumours include a subgroup with aggressive disease, and co-mutation with *STK11* is an obvious candidate [2], but with 39 *STK11*-mutated patients this cohort is too small to test an interaction.

The expression analysis suggests why a binary indicator performs poorly. The same mutations that add nothing to prognosis are clearly visible at the RNA level, and the direction of the differences matches what is known about these genes. Activating *EGFR* and *KRAS* mutations frequently coexist with amplification of the mutant allele, which raises transcript abundance. Many *STK11* and *TP53* variants are truncating and expose the transcript to nonsense-mediated decay, whereas *TP53* missense variants usually leave mRNA levels unchanged, which would explain the smaller and noisier difference for that gene. A single yes/no flag pools these different classes of variant and discards the quantitative information that the expression layer retains. A natural next step is to replace the four flags with variant class, co-mutation patterns and expression-derived features, and to test them with the same optimism-corrected framework used here.

Limitations

Restricting the cohort to patients with a day-zero sample removed 40% of otherwise eligible cases. The restriction gives a cleaner time origin but costs power, and the selected patients may differ from the rest; repeating the analysis on all 489 eligible patients is the first sensitivity analysis I would add. Validation was internal only: the corrected C-index estimates performance in patients like these, and an independent cohort would be needed to confirm it. I did not compute a confidence interval for the corrected ΔC. The proportional-hazards assumption is questionable for *KRAS*, and I did not fit a time-varying or stratified model. TCGA records limited treatment information, so the models cannot separate prognostic effects from effects of therapy, which is relevant for *EGFR* in particular. Follow-up is short (median 30.8 months), and estimates beyond five years rest on few patients. Finally, the expression comparisons are unadjusted for tumour purity and copy number and should be read as descriptive.

5 Conclusion

In 295 patients with lung adenocarcinoma, adding the mutation status of *EGFR*, *KRAS*, *TP53* and *STK11* to age, sex and stage produced no meaningful improvement in survival discrimination after correction for optimism. The mutations are nonetheless reflected in the expression of their own genes, which points to richer representations of the genomic and transcriptomic layers as the place to look for prognostic value. The analysis is fully reproducible from public data.

Data and code availability

All data come from the public TCGA-LUAD PanCancer Atlas study on cBioPortal. The repository accompanying this report contains the analysis table, the scripts that rebuild the cohort from the source files, the survival notebook, the expression analysis, and the script that generates every figure and number

reported here (`scripts/make_report_figures.py`).

Acknowledgements

The results shown here are based upon data generated by the TCGA Research Network (`https://www.cancer.gov/tcga`). I thank the patients who contributed samples and the cBioPortal team for maintaining open access to the data.

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